Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract

This report evaluates `minimax/minimax-m2.7` under the standard Infron vs OpenRouter Prompt Caching A/B benchmark. Token cache hit rate: Infron leads in 2/4 routing modes; Observed cost: Infron leads in 4/4 routing modes; Throughput: OpenRouter leads in 3/4 routing modes; E2E latency: OpenRouter leads in 3/4 routing modes; TTFT: OpenRouter leads in 3/4 routing modes. The practical interpretation is that platform choice should be tied to the target objective instead of a single global metric.

The A/B filter retained 800 paired samples and 3200 request-level observations; 0 records were excluded. All core metrics come from response telemetry.

Conclusion Matrix

Routing mode	Cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	**Infron**	**Infron**	**OpenRouter**	**OpenRouter**	**OpenRouter**
Price First	**Infron**	**Infron**	**Infron**	**Infron**	**Infron**
Latency First	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	**OpenRouter**
TTFT First	**OpenRouter**	**Infron**	**OpenRouter**	**OpenRouter**	**OpenRouter**

Overall Metrics

Routing mode	Platform	Pairs	Token cache hit rate	Actual cost	Avg throughput	Avg E2E latency	Avg TTFT
Throughput First	Infron	200	**97.41%**	**$0.36936000**	2.67 tok/s	5980.69 ms	5786.47 ms
Throughput First	OpenRouter	200	93.54%	$3.59489664	**3.39 tok/s**	**4723.73 ms**	**4423.83 ms**
Price First	Infron	200	**98.58%**	**$0.35053300**	**2.60 tok/s**	**6150.60 ms**	**5959.01 ms**
Price First	OpenRouter	200	66.56%	$0.72619654	1.79 tok/s	8930.26 ms	8499.25 ms
Latency First	Infron	200	98.43%	**$0.39538500**	5.30 tok/s	3016.90 ms	2837.35 ms
Latency First	OpenRouter	200	**98.90%**	$3.53757760	**5.49 tok/s**	**2916.05 ms**	**2564.25 ms**

Routing mode	Platform	Pairs	Token cache hit rate	Actual cost	Avg throughput	Avg E2E latency	Avg TTFT
TTFT First	Infron	200	98.43%	**$0.37421000**	4.88 tok/s	3281.74 ms	2994.02 ms
TTFT First	OpenRouter	200	**98.88%**	$3.63688800	**5.56 tok/s**	**2875.68 ms**	**2440.43 ms**

Prompt Length Stratification

Tier	Platform	Pairs	Token cache hit rate	Actual cost	Avg E2E latency	Avg TTFT
short (1500 target tokens)	Infron	268	77.71%	**$0.13344700**	3632.64 ms	3421.09 ms
short (1500 target tokens)	OpenRouter	268	**81.22%**	$0.46397534	**3568.51 ms**	**3223.26 ms**
medium (8000 target tokens)	Infron	268	**98.91%**	**$0.29008400**	**4367.24 ms**	**4173.93 ms**
medium (8000 target tokens)	OpenRouter	268	87.49%	$2.26869193	4592.34 ms	4219.52 ms
long (32000 target tokens)	Infron	264	**99.05%**	**$1.06595700**	**5840.97 ms**	**5605.71 ms**
long (32000 target tokens)	OpenRouter	264	90.38%	$8.76289151	6447.10 ms	6026.10 ms

Method And Reproducibility

The run used the standard 4 groups x 50 rounds design, streaming Chat Completions, platform-default reasoning/thinking behavior, prompt-length tiers `short:1500,medium:8000,long:32000`, and `/v1/chat/completions` only. A/B inclusion allowed up to 50 prompt-token difference within each paired first/second request.

Artifact	Path
Summary JSON	`export/minimax_m27_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_summary.json`
Paired dataset CSV	`export/minimax_m27_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_benchmark_pairs.csv`
Request-level JSONL	`export/minimax_m27_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_benchmark_requests.jsonl`
Filtered records	`export/minimax_m27_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records.json`
Excluded-record audit	`export/minimax_m27_all_experiments/routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions_records_excluded.json`

Artifact	Path
Benchmark runner source	`scripts/rerun_routing_sort_cache_cost_ab.py`
HTML report renderer source	`scripts/render_glm52_deepseek_style_report.py`
Test source	`tests/test_rerun_routing_sort_cache_cost_ab.py`
Dataset reference	Built-in `controlled_cache_probe` with prompt-length tier construction; request-level export in `benchmark_request`